

Review of the corrected AR(1) Section 5.3 study

SVP simulation audit

2026-09-15

Study summary

The corrected study uses stationary Gaussian AR(1) noise with $n = 600$, $\rho = 0.8$, unit marginal variance, four patterns, jumps from 1 to 5 in steps of 0.2, 100 replications, and a five-observation matching tolerance. The final result contains 33600 method rows and the four methods PELT AR1 approximate, PELT inflated, DeCAFS AR1, SVP AR1Focus / right / c = 1.850.

The mandatory endpoint n is removed before one-to-one metric matching. The reported metrics are precision, recall, F1, correct-number probability, signal MSE, fitted segment count, and localization error.

Calibration

The selected DeCAFS multiplier is 7.75 and its calibration null F1 is 0.99. The selected approximate-Pelt multiplier is 0.8; the selected AR1Focus multiplier is 1.85.

DeCAFS is the calibration target at corrected null F1 0.99. The approximate Pelt and AR1Focus candidates are selected by closest calibration null F1 to that DeCAFS target. The inflated PELT method remains a fixed paper baseline. Calibration and validation use 1,000 independent null series each, with deterministic seeds and Wilson intervals.

Table 1: Validation null F1 for the selected methods.

method	candidate	null_f1	null_f1_lower	null_f1_upper
DeCAFS AR1	7.75	0.985	0.9754	0.9909
PELT AR1 approximate	0.80	0.989	0.9804	0.9938
SVP AR1Focus	1.85	0.993	0.9856	0.9966
PELT inflated	3.00	0.999	0.9944	0.9998

PELT approximation and AR1Focus

The approximate AR(1) PELT method applies the package PELT implementation to $Z_t = Y_{t+1} - \rho Y_t$, then maps an endpoint j in the transformed series to endpoint $j+1$ in the original series. This is an approximation because the complete AR(1) likelihood includes a boundary correction that is not retained by ordinary transformed-data least squares.

AR1Focus is called with `test = "AR1Focus"` and `subtests = "right"`. No `both` or `multiscale` SVP call is included in the study.

Legacy comparison

Table 2: Corrected AR(1) study summary.

algorithm	F1	Precision	Recall	CorrectNumCP	MSE	NumSegments
DeCAFS AR1	0.7244	0.8317	0.6775	0.4638	0.4414	4.0714
PELT AR1 approximate	0.6744	0.6352	0.7744	0.2760	0.2435	7.7911
PELT inflated	0.7485	0.8319	0.7061	0.5033	0.3524	4.4107
SVP AR1Focus	0.7969	0.8706	0.7578	0.5469	0.2667	4.8712

Table 3: Archived pre-refresh AR(1) study summary.

algorithm	F1	Precision	Recall	CorrectNumCP	MSE	NumSegments
DeCAFS AR1	0.7862	0.8729	0.7435	0.5230	0.2694	4.6460
PELT AR1 approximate	0.5975	0.6367	0.6070	0.2948	0.6641	4.7062
PELT inflated	0.7485	0.8319	0.7061	0.5033	0.3524	4.4107
SVP AR1Focus	0.6700	0.7711	0.6241	0.4161	0.6211	3.9121

Reproduction

```
source("simulations/power_ar1/calibrate_ar1.R")
calibration <- run_ar1_calibration(workers = 8L)
source("simulations/power_ar1/run_power.R")
run_and_save_ar1(workers = 8L, calibration = calibration)
source("simulations/power_ar1/export_paper_figures.R")
export_ar1_paper_figures(calibration = calibration)
```